PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

CampusPulse: Geo-Location Based Attendance System Using Wi-Fi BSSID Validation and Multi-Factor Authentication

A PROJECT REPORT

Submitted by

SUMITH KUMAR V- 20221COM0192

RAVI RANJAN-20221COM0185

HRISHIKESH VIVEKANAND GHODKE-20221COM0152

Under the guidance of,

Dr. Asha Hegde

(Assistant Professor)

BACHELOR OF TECHNOLOGY

IN

COMPUTER ENGINEERING

PRESIDENCY UNIVERSITY

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BONAFIDE CERTIFICATE

Certified that this report **"CampusPulse: Geo-Location Based Attendance System Using Wi-Fi BSSID Validation and Multi-Factor Authentication"** is a bonafide work of **"SUMITH KUMAR V (20221COM0192),RAVI RANJAN(20221COM0185), HRISHIKESH VIVEKANAND GHODKE (20221COM0152)"**, who have successfully carried out the project report for the partial fulfilment of the requirements for the award of the degree **BACHELOR OF TECHNOLOGY** in **COMPUTER ENGINEERING**, during **2025-2026.**

Dr. Asha Hegde
Guide, Assistant Professor
PSCS,
Presidency University

24/4/26.
Ms. Sumita Guddin
Program Project Coordinator
PSCS,
Presidency University

25/4/26
Dr. Gopal Krishna Shyam,
Head of the Department,
PSCS,
Presidency University

Dr. Duraipandian N,
Dean,
PSCS,
Presidency University

Name and Signature of the Examiners

Sl. No	Name	Signature	Date
1	Dr. Asha Hegde		24/04/2026
2	Jubin Jacob		24/4/26

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DECLARATION

We the students of final year **B.Tech in Computer Engineering at Presidency University, Bengaluru**, named **Sumith Kumar V, Ravi Ranjan, Hrishikesh Vivekanand Ghodke,** hereby declare that the project work titled "**CampusPulse: Geo-Location Based Attendance System Using Wi-Fi BSSID Validation and Multi-Factor Authentication"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree "**Bachelor of Technology in Computer Engineering",** during the academic year of **2025-26**. Further the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Sumith Kumar V	Usn : 20221COM0192	
Ravi Ranjan	Usn : 20221COM0185	
Hrishikesh Vivekanand Ghodke	Usn : 20221COM0152	

PLACE: BENGALURU

DATE: 24-April 2026

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SUMITH KUMAR V
RAVI RANJAN
HRISHIKESH VIVEKANAND GHODKE

Abstract

Institutions of learning are continually challenged by attendance management problems. Some examples include manual processes that require a lot of time, proxy attendance, late updates to records, and inconsistent attendee verification processes. While existing systems using mobile phones and GPS-based location technologies increase the speed of administrative functions, the majority of current deployments still use low-security single-factor authentication methods or rely on manipulated location signals that can be spoofed. Regarding indoor education environments, Wi-Fi networks present a reasonable, low-cost solution for providing context to the verification of attendance; however, previous studies have shown that using identification-based location data may expose the potential for privacy breaches if sufficient precautions are not taken with that data. These issues indicate that there is an ongoing need for an automated attendance system that is not only secure and dependable but also supports routine operational use by institutions.

This project report describes a smartphone-based attendance solution that uses WiFi BSSID information as proximity inputs and combines this information with time-based validation, role-based control, and tiered identity assurance. The design of the system includes unique workflows for administrator, faculty, and student system users. Attendance is only recorded if it occurs during a specific time slot and from a network context consistent with the expected class. This framework will apply biometrics and multi-factor authentication to see an improvement to the legitimacy of each attendance event by limiting the use of password-only access and providing a measure of resistance to impersonation. Additionally, abnormal check-ins will be identified using anti-spoofing concepts taken from GNSS security literature allowing for discrepancies in check-in location and session patterns to be identified as suspicious.

The model offered provides a practical implementation of a secure attendance monitoring system in campus environments without requiring expensive hardware upgrades. This will reduce instructor administrative burden, improve audit tracking of records, and improve confidence in presence verification compared to traditional manual and/or weakly validated forms of digital verification of presence. Overall this project shows that combining Wi-Fi context, stronger methods of authentication, and anomaly aware check-ins provides a scalable (due to current infrastructure), cost-effective, and trustworthy attendance management solution to higher education institutions.